# Build the full prediction

*Mission control wants force → acceleration → velocity → position.*

20 A

**THE SCENARIO**

A 4 kg cart starts from rest at x = 2 m. A 14 N pull acts right; friction is 2 N left. The forces act for 3 s.

**LINK 1 · NET FORCE**

$\Sigma F =$ ____ N

**LINK 2 · ACCELERATION**

$a = \Sigma F \div m =$ ____

**LINK 3 · VELOCITY**

$v = v_0 + at =$ ____

**LINK 4 · POSITION**

$x = x_0 + \frac{1}{2}(v_0 + v)t =$ ____

**RANGE**

The model bound is ±1.5 m. Report: ____ m to ____ m. Guaranteed? ____

**WHAT IF…**

…friction rose to 14 N? Which link changes first and what happens?

**ANALYST SIGN-OFF**

The link I trust least is ____ because ____. For 2026-XJ, we do the east and north predictions separately because ____.

SYNTHESIS 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

VOL. 1 · NO. 20 · SYNTHESIS REPORTER: _____

# The 2026-XJ Daily

**Cart prediction confirmed — or was it?**

**BREAKING · HEADLINE (include the range)**

**THE STORY**

Explain how the team got from forces to a final position. Name each link in the chain.

**BY THE NUMBERS**

ΣF = ____

a = ____

v = ____

x = ____

range ____

**LETTERS TO THE EDITOR**

"The cart has zero net force, so it can't move." — Confused in Room 108. The editor replies:

**SCIENCE DESK**

Zero net force, already moving at +2 m/s for 3 s: velocity and displacement?

**CORRECTION BOX**

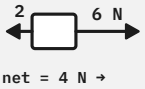
Yesterday we printed "5 m — guaranteed." Write the correction:



UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Picture dictionary: the chain

Language goal: I can **connect** cause and effect.



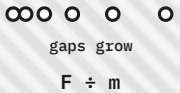

## net force

fuerza neta · força resultante

all forces added up

my language: _____

use it: _____



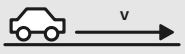

## acceleration a

aceleración · aceleração

how velocity changes

my language: _____

use it: _____




$v_0 + at$

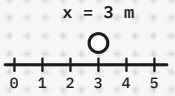
## velocity v

velocidad · velocidade

how fast + which way

my language: _____

use it: _____




## position x

posición · posição

where it is

my language: _____

use it: _____




## cause

causa · causa

what makes something happen

my language: _____

use it: _____




## effect

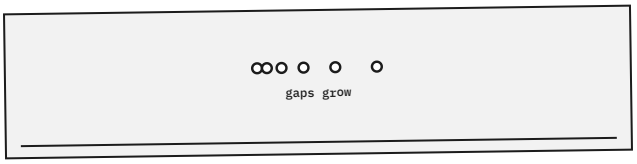
efecto · efeito

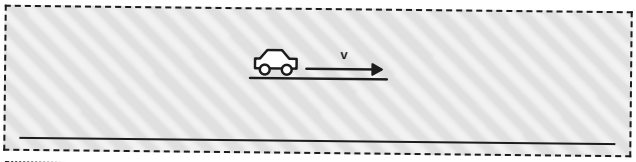
what happens because of it

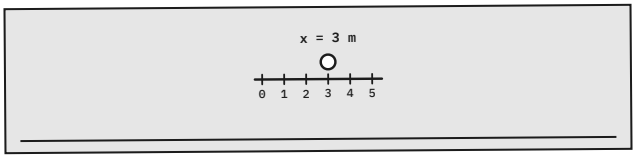
my language: _____

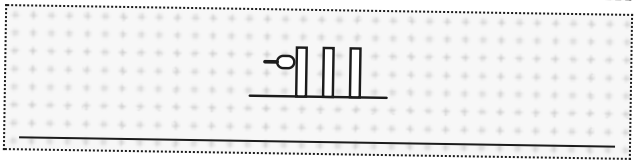
use it: _____

**MATCH · WRITE THE WORD FOR EACH PICTURE**



I can point and say all 6 words to a partner □

SYNTHESIS (1) (2) (3)

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

ANTOCCI ENGINEERING · DWG U1-20 · SYNTHESIS

# DELIVERY DRONE: LANDING SEQUENCE

A 2 kg drone descends. Up = +. Plan the chain from forces to landing.

### CALC 1 · NET FORCE

Thrust 16 N up, weight 19.6 N down. $\Sigma F$ = ____

### CALC 2 · ACCELERATION

$a = \Sigma F \div m$ = ____ m/s²

### CALC 3 · VELOCITY AFTER 2 s

Starts at 0 m/s. $v = v_0 + at$ = ____

### CALC 4 · DROP IN 2 s

$\Delta y = \frac{1}{2}(v_0 + v)t$ = ____

### DRAWING · FBD + v–t SKETCH OF THE DESCENT

### ENGINEER'S NOTE

What thrust would make the drone descend at a constant velocity? Explain with the 1st law.

SYNTHESIS 1 2 3

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☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 20 · SYNTHESIS · SEI · SENTENCE FRAMES

# Say it, then write it: the chain

Language goal: I can **sequence** cause and effect.



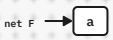
**cause**
causa · causa
what makes it happen



**effect**
efecto · efeito
what happens




**then**
luego · então
next in order




**so**
así que · então
shows a result

English · Spanish · Portuguese. My home language: _____

## 1 · READ AND FILL · WORD BANK: force · acceleration · velocity · position · range

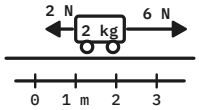
A net ____ causes an ____. Then the ____ changes. So the ____ changes too. We report a ____ because the model is not perfect.

## 2 · SAY IT TO A PARTNER, THEN WRITE IT




The net force is ____, so the acceleration is ____ .
Then the velocity is ____, so the position is ____ .

## 3 · SOLVE WITH THE STEPS




2 kg cart, 6 N right, 2 N left, from rest at 1 m, for 2 s.

| STEP 1 · ΣF AND a | STEP 2 · v | STEP 3 · x |
|---|---|---|
| ____ N · ____ m/s² | 0 + ____ × ____ = ____ | 1 + ____ = ____ m |

## 4 · EXPLAIN IT · CHOOSE A FRAME

☐ First ____, then ____, so ____ .
☐ The cause is ____ and the effect is ____ .
☐ We give a range because ____ .

I said it ☐ · I wrote it ☐

SYNTHESIS 1 2 3

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# Fix the group chat

20 R

**DIRECTIONS** · Reply in the empty bubbles. Show the math on the right.

### XJ-7 tracking crew

**NADIA**

2 kg cart, 6 N pull, 2 N friction, from rest. a?

**YOU ①**

**ELI**

cool so after 2 s it's at 4 m/s

**MAZ**

how far did it go in those 2 s?

**YOU ②**

**ELI**

it started at 1 m. so it's at exactly 5 m, guaranteed

**YOU ③**

**NADIA**

and if the net force were zero?

**YOU ④**

**SHOW YOUR RECEIPTS**

① ΣF → a

② Δx = avg v × t

③ $x_0$ + Δx, with a range

④ ΣF = 0 →

**SYNTHESIS** 1 2 3

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Comic: from force to "where will it be?"

20 CS

**YOUR READER**

A younger student who asks, "How can scientists know where an asteroid will be?"

Stick figures are fine. The physics gets checked.

**MUST-HAVES · check it off, then color its circle**

- ☐ ◌ **ORANGE** — The force arrow in panel 1
- ☐ ◌ **G/R** — Green if it speeds up, red if it slows down
- ☐ ◌ **PURPLE** — Velocity arrows in panel 2
- ☐ ◌ **BLUE** — The predicted position in panel 3

**1 · FORCE → ACCELERATION** *A force acts on 2026-XJ (or a cart).*

**CAPTION** _____

**2 · ACCELERATION → VELOCITY** *Its velocity changes over time.*

**CAPTION** _____

**3 · VELOCITY → POSITION** *Predict where it will be.*

**CAPTION** _____

**EXPLAIN IT ·** In 1–2 sentences, what physics does your comic show? Use: **force · acceleration · velocity · position · predict**

_____

_____

**SYNTHESIS** 1 2 3

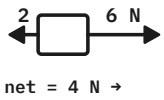
**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Doodle Notes · net force

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

**Net force** · fuerza neta (ES) · força resultante (PT)




**TODAY'S CASE**

A 2 kg cart starts from rest with a constant 4 N net force right for 2 s.

## 01 · SKETCH AND LABEL

Draw and label the orange net force. Sketch how the purple velocity arrow changes each second.

## 02 · CHANGE THE PICTURE

Use your arrows to sketch position dots at equal times. Explain what the dots would do if the net force became zero at 2 s.

**Annotate your sketches.** Circle the mark that best shows net force. Add a label in English or your home language.

**SAY IT TO A PARTNER · THEN WRITE**

The force changes ___, so the spacing …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

# Frayer Model · net force

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

**Net force** · fuerza neta (ES) · força resultante (PT)

**A** · A constant net force changes a cart's velocity.

**B** · A force is always needed to maintain an existing velocity.

### 01 · DEFINITION

What does net force mean in your own words? Use the notes or start in your home language.

### 02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

### 03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

### 04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

**SAY IT TO A PARTNER · THEN WRITE**

I chose card ___ as my example because …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else